

CLONE: Synthetic Guideline-based Clinical Reasoning with Large Language Models for Early Diagnosis of Mild Cognitive Impairment

Seungeon Cha
Seoul National University of Science
and Technology
Seoul, Republic of Korea
secha@seoultech.ac.kr

Jinseok Park
Hanyang University
Seoul, Republic of Korea
jinspark@hanyang.ac.kr

Hojin Choi
Hanyang University
Seoul, Republic of Korea
chj@hanyang.ac.kr

Hokyoung Ryu
Hanyang University
Seoul, Republic of Korea
hryu@hanyang.ac.kr

Kyoungwon Seo
Seoul National University of Science
and Technology
Seoul, Republic of Korea
kwseo@seoultech.ac.kr

Abstract

Early diagnosis of mild cognitive impairment (MCI) is essential to prevent its progression to Alzheimer’s disease. Human expert-driven diagnosis provides interpretable rationales but is time-consuming, while machine learning-based approaches offer efficiency but lack human-readable rationales. To address these limitations, we propose **CLONE** (Clinical Reasoning via Neuropsychologist Emulation), a three-stage framework leveraging large language models (LLMs) for MCI diagnosis: (1) emulating experts through role-playing, (2) synthesizing step-by-step diagnostic guidelines, and (3) performing clinical reasoning using the guideline. CLONE was evaluated on a real-world dataset of 65 subjects, achieving 89.23% diagnostic accuracy and outperforming the few-shot chain-of-thought (CoT) baseline by 6.15%, with specificity improving by 10.71%. Moreover, the synthesized guideline enhanced rationale quality, making rationales more consistent, correct, specific, helpful, and human-like compared to baselines. These findings highlight CLONE’s potential to enable accurate diagnosis and reliable clinical reasoning, addressing challenges in the field of MCI diagnosis. Our code is available at <https://github.com/seoultech-HAILAB/CLONE>.

CCS Concepts

• Computing methodologies → Natural language generation.

Keywords

Interpretable AI Diagnosis, Mild Cognitive Impairment, Large Language Models, Clinical Reasoning, Synthetic Guidelines

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1 Introduction

Early diagnosis of mild cognitive impairment (MCI), a transitional stage between normal aging and Alzheimer’s disease (AD) [6], is critical for preventing the progression to AD through timely intervention [26]. MCI represents a pivotal stage where effective diagnosis and management can significantly delay the onset of the irreversible stages of AD, which severely impacts a patient’s ability to perform daily activities independently [27]. Over the years, various biomarkers have been developed for MCI diagnosis, including neuropsychological tests [2, 29], magnetic resonance imaging (MRI) [3, 34], cerebrospinal fluid (CSF) analysis [28], and others [9, 17]. Among these, neuropsychological tests are particularly valuable in the initial diagnostic stage due to their non-invasive nature and lower cost compared to other methods [11].

A key component of neuropsychological testing is the rationale provided by human experts, which includes interpreting test scores, outlining the reasoning path to a diagnosis, and delivering a clinical decision (e.g., whether to conduct further testing or initiate interventions) [24]. These rationales, written in natural language, provide critical insights into a subject’s cognitive status and guide clinical decision-making [18]. However, as illustrated in Figure 1, this human expert-driven process is time-intensive, requiring neuropsychologists to produce comprehensive reports that integrate interpretations, reasoning, and diagnostic conclusions. Automated diagnostic methods using machine learning (ML) have the potential to address some of these challenges by offering faster and more cost-efficient diagnoses [23]. However, they often lack the ability to provide interpretable, human-readable rationales, limiting their utility in clinical contexts where transparency is essential [15, 25]. Consequently, there is a growing need for diagnostic approaches that combine the efficiency of ML with the interpretability of expert-driven reasoning.

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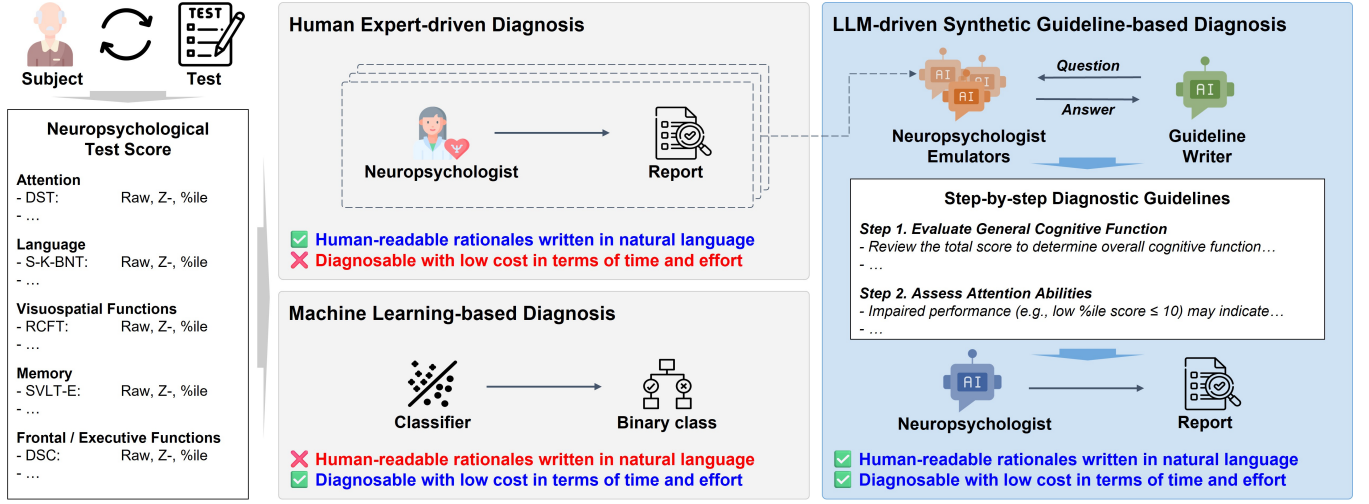


Figure 1: Comparison of methods for MCI diagnosis. Human expert-driven diagnosis provides interpretable rationales but are resource-intensive (gray box at the top), while machine learning-based diagnosis is cost-efficient but cannot offer human-readable rationales (gray box at the bottom). In contrast, our proposed LLM-driven synthetic guideline-based diagnosis method offers both rationales written in natural language and automated diagnosis with low cost (blue box).

Recent advances in large language models (LLMs) offer a promising solution to these challenges. Leveraging LLMs’ natural language generation capabilities enables the generation of human-readable rationales alongside diagnostic predictions [22, 32, 33]. Moreover, LLMs have demonstrated reasoning capabilities comparable to medical experts by “thinking before answering,” enhancing both diagnostic accuracy and interpretability [8, 19, 30]. Inspired by these advancements, this study introduces **CLONE** (Clinical Reasoning via Neuropsychologist Emulation), a novel framework designed to improve diagnostic accuracy while generating interpretable rationales. This research seeks to address the following questions:

- **RQ1:** How does an LLM-driven synthetic guideline affect diagnostic accuracy through clinical reasoning?
- **RQ2:** How does an LLM-driven synthetic guideline impact on the quality of generated rationales?

The remainder of this paper is structured as follows: Section 2 reviews the background on MCI diagnosis and LLM-based clinical reasoning. Section 3 describes the proposed CLONE framework. Section 4 details the experimental setup, including dataset and evaluation metrics. Section 5 presents the experimental results. Finally, Section 6 discusses the findings and outlines future research directions.

2 Background

2.1 Early Diagnosis of Mild Cognitive Impairment

For the early diagnosis of MCI, a variety of biomarkers have been developed and utilized including neuropsychological tests, imaging techniques, and molecular analyses [7, 20]. Among these, neuropsychological tests stand out as the most practical and user-friendly due to their non-invasive nature, accessibility, and cost-effectiveness compared to other biomarkers [11]. A representative example is

the Seoul Neuropsychological Screening Battery-Core (SNSB-C), which has been widely employed for assessing cognitive function in clinical settings [12]. These tests not only serve as an effective first-line diagnostic tool but also provide a detailed assessment of cognitive domains that may be affected during the transition from normal aging to MCI.

Neuropsychological tests can be analyzed either by human experts or through ML methodologies, each offering distinct advantages and drawbacks. When analyzed by human experts, such as neuropsychologists, these tests provide interpretable and context-aware insights into a subject’s cognitive status [18]. However, this process is often time-consuming and labor-intensive [24]. On the other hand, ML-based methods offer automated and cost-efficient analysis, capable of leveraging structured test scores to train predictive models that detect patterns indicative of MCI. While ML approaches excel in scalability and speed, they often lack interpretability and the nuanced reasoning required for clinical use [15, 23]. Recently, LLMs have shown promise in addressing these limitations [22, 32]. By incorporating the reasoning capabilities of human experts with the efficiency of ML, LLMs can complement the strengths of both approaches [30].

2.2 LLM-based Clinical Reasoning in Medical Domain

Recent advancements in LLMs have demonstrated their potential to conduct clinical reasoning [19]. LLM-based clinical reasoning follows a sequential process where patient information I (e.g., demographic data or neuropsychological test scores) is provided as input. The LLM sequentially generates rationales R and final diagnosis D , as described below:

$$R = LLM(I) \rightarrow D = LLM(I, R)$$

This sequential generation of tokens allows LLMs to enable the reasoning process like humans. Recent studies have shown that LLMs can produce results closely aligned with human reasoning in complex medical scenarios [8]. For instance, Kwon and colleagues [19] showed that clinical reasoning of acceptable quality can be achieved even with smaller models by fine-tuning them through knowledge distillation. Additionally, multi-agent frameworks such as MedAgents [35] and MDAgents [16] have demonstrated that collaboration among agents acting as medical experts can facilitate accurate diagnoses based on robust clinical reasoning.

However, despite these advancements, a critical gap remains in effectively combining the interpretability of human expert reasoning with the efficiency and scalability of automated systems. Human experts provide nuanced, context-aware rationales but are constrained by the time and effort required for analysis [18, 24]. Machine learning methods, while efficient, often lack transparency and the ability to justify their conclusions [15, 23]. LLMs have the unique potential to address this gap by synthesizing diagnostic guidelines that emulate expert reasoning and producing reliable rationales alongside accurate diagnosis [8, 19, 30]. Such capabilities can bridge the divide between interpretability and automation, enabling more effective diagnostic processes in clinical practice.

Therefore, inspired by these possibilities, this study proposes a novel framework to leverage LLMs for synthesizing diagnostic guidelines through the emulation of human expert reasoning (see Figure 1). The framework aims to address existing limitations by generating interpretable, human-readable rationales while ensuring diagnostic accuracy, paving the way for more transparent and efficient diagnostic tools in the medical domain.

3 The Proposed Framework: CLONE

The proposed framework, CLONE, is designed to emulate expert clinical reasoning, synthesize diagnostic guidelines, and apply them for accurate and interpretable diagnoses. CLONE comprises three stages: (1) emulating experts through role-playing (Section 3.1), (2) synthesizing step-by-step diagnostic guidelines (Section 3.2), and (3) performing clinical reasoning using the synthesized guideline (Section 3.3). An overview of the CLONE framework is illustrated in Figure 2, and detailed pseudo-code is provided in Appendix A (see Algorithm 1).

3.1 Stage 1: Emulating Experts through Role-Playing

The first stage aims to uncover the reasoning process of human neuropsychologists by simulating their interpretation of neuropsychological test scores and diagnoses (i.e., underlying justifications). CLONE samples a small subset $D_{sub} = \{(s_j, r_j) | r_j = p(s_j)\}_{j=1}^n$, where $(n < N)$, from dataset $D = \{(s_i, r_i) | r_i = p(s_i)\}_{i=1}^N$. Here, $S = \{s_1, \dots, s_N\}$ represents neuropsychological test scores for N human subjects, and $R = \{r_1, \dots, r_N\}$ represents their corresponding diagnostic reports, encompassing the clinical reasoning process leading to the diagnosis, as well as the final diagnosis, written by M neuropsychologists $P = \{p_1, \dots, p_m\}$.

Each sample in D_{sub} is provided to n emulator agents $A_e = \{a_e^1, \dots, a_e^n\}$ to enable each agent a_e^j to mimic the clinical reasoning process of a neuropsychologist p_j . A guideline writer agent A_g

interacts with the emulator agents A_e by posing question set Q (e.g., “How did you interpret the test results?” and “What were the decisive factors for the diagnosis?”). The emulator agent a_e^j respond with detailed answer $\hat{a}_j$, describing their reasoning processes. The responses from all n agents are collected into an answer set $\hat{A} = \{\hat{a}_1, \dots, \hat{a}_n\} = A_e(S, R, Q)$. This role-playing process is executed in parallel with all agents to minimize latency. Examples of prompts and outputs from this stage are provided in Appendix B.1 (see Table 2).

3.2 Stage 2: Synthesizing Step-by-step Diagnostic Guidelines

In the second stage, the collected responses $\hat{A}$ are synthesized into a step-by-step diagnostic guideline G , which outlines the interpretation methods for neuropsychological test scores and reasoning strategies for diagnosis. The guideline writer agent A_g drafts preliminary guidelines G_{draft} for each response in $\hat{A}$:

$$G_{draft} = g_1, \dots, g_n = A_g(\hat{A})$$

These preliminary drafts are then consolidated by A_g into a unified, coherent guideline G :

$$G = A_g(G_{draft})$$

This synthesis process ensures that the guideline G captures the diverse perspectives and reasoning strategies observed during role-playing while maintaining clarity and coherence. Similar to the first stage, this process is parallelized for efficiency. Examples of prompts and outputs from this stage are provided in Appendix B.2 (see Table 3).

3.3 Stage 3: Clinical Reasoning using the Guideline

The final stage applies the synthesized guideline G to perform clinical reasoning and diagnose MCI for a target subject. If a guideline G already exists, the first two stages can be skipped. Given the neuropsychological test scores s^* of a target subject, the neuropsychologist agent A_p generated a rationale R^* and diagnosis D^* sequentially using the guideline G as below:

$$R^* = A_p(G, s^*) \rightarrow D^* = A_p(G, s^*, R^*)$$

This sequential generation process ensures interpretability and accuracy in the diagnosis. Examples of clinical reasoning and diagnostic outputs are provided in Appendix B.3 (see Table 4).

4 Experimental Design and Evaluation

4.1 Dataset

To construct the real-world dataset D for this study, we recruited 65 subjects from Hanyang University Hospital in Seoul. The dataset consisted of 28 healthy controls (HC), labeled as ‘0’ (negative group), and 37 individuals diagnosed with MCI, labeled as ‘1’ (positive group). All diagnoses were determined by consensus between two neurologists with more than 10 years of experience, ensuring reliability. The neuropsychological assessments were conducted using the SNSB-C, a standardized tool for the Korean population [12]. This battery evaluates five key cognitive domains—*attention, language*

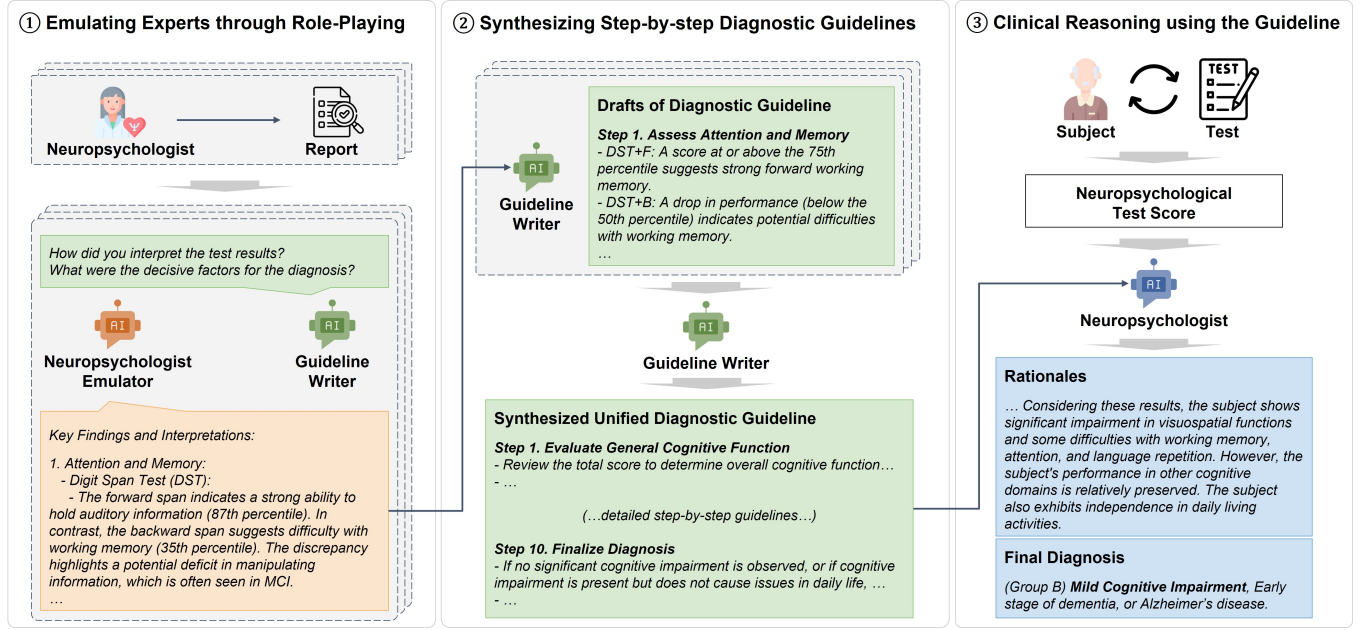


Figure 2: Overview of CLONE for clinical reasoning using synthesized diagnostic guidelines. Given a small subset of reports authored by human neuropsychologists, CLONE models their diagnosis process through role-playing (i.e., question and answer) between agents emulating these experts and a guideline writer agent (①). Using the resulting set of responses, the guideline writer agent constructs a unified step-by-step diagnostic guideline (②). This synthesized guideline is then used by a neuropsychologist agent to conduct diagnosis through clinical reasoning tailored to the target subject (③).

function, visuospatial function, memory, and frontal/executive functions—through dedicated subtests. All assessments were conducted by a licensed neuropsychologist with over 15 years of experience, ensuring high-quality and consistent data collection.

4.2 Experimental Setup

We evaluated only open-source LLMs that can be deployed locally, considering the sensitive nature of medical data and the need to prioritize subjects' data privacy. To assess the effectiveness of the proposed CLONE framework, we utilized state-of-the-art open-source LLMs of varying sizes, including Llama 3.3 (70B), Phi 4 (14B), Llama 3.1 (8B) [1, 4], and DeepSeek-R1-Distill series [10].

To ensure a balanced and controlled evaluation, a subset D_{sub} comprising 10% of the dataset (6 subjects: 3 HC and 3 MCI) was randomly selected for guideline development. This small yet representative subset was used to synthesize the diagnostic guideline, which was subsequently employed during the evaluation phase. In other words, the evaluation was conducted on the full dataset D , consisting of 65 subjects, including the 6 samples used for guideline synthesis, ensuring consistency and reliability.

The performance of the LLMs was assessed under three configurations: zero-shot chain-of-thought (CoT) [36], few-shot CoT, and the proposed CLONE framework. In the zero-shot CoT setup, no prior examples were provided to guide the models. For the few-shot configuration, examples were randomly sampled from the dataset while ensuring a balanced selection between HC and MCI groups

(e.g., for 2-shot, 1 HC and 1 MCI example were chosen) to minimize potential biases. The CLONE setup leveraged the synthesized unified guideline to enhance both diagnostic accuracy and interpretability. Across all configurations, the temperature parameter was fixed at 0.1 to maintain.

4.3 Evaluation

To evaluate the effectiveness of the proposed CLONE framework, we focused on two key aspects: the accuracy of diagnosis guided by the synthesized guideline and the quality of the generated rationales. These two aspects were selected to comprehensively evaluate both the performance and interpretability of the framework.

For the first aspect, we assessed the binary classification performance of the models in distinguishing between HC and individuals with MCI. To achieve this, standard metrics such as accuracy, sensitivity, specificity, precision, and F1-score were employed [21]. These metrics provided a detailed understanding of the models' diagnostic capabilities, measuring their ability to correctly classify both positive (MCI) and negative (HC) cases.

The second aspect focused on the quality of the rationales generated by the models during the diagnostic process. To evaluate rationale quality, we employed the LLM-as-Judge rubric evaluation method [37], a reliable approach that has been validated for its consistency with human judgment [13]. Specifically, we utilized Prometheus 2 (8x7B) [14] as the evaluator, comparing rationales generated under different configurations. The evaluation was based on a predefined rubric that assessed five key criteria: *Consistency*,

absence of contradictions between the presented data, rationale, and prediction; *Correctness*, absence of misconceptions in the medical knowledge used in the rationale; *Specificity*, level of detail of the insights provided in the rationale; *Helpfulness*, contribution of the rationale to achieving an accurate diagnosis; and *Human-likeness*, similarity to explanations typically provided by human experts. This rubric was adapted from prior study [19] and is described in detail in Appendix C (see Table 5).

5 Results

5.1 RQ1: How does an LLM-driven synthetic guideline affect diagnostic accuracy through clinical reasoning?

Table 1 summarizes the diagnostic performance of the proposed CLONE framework compared to zero-shot and few-shot CoT baselines across different LLMs. The results highlight CLONE’s ability to improve diagnostic accuracy and other related metrics in general-purpose models such as Llama 3.3 (70B) and Phi 4 (14B), while this improvement was not observed in smaller model (e.g., Llama 3.1 (8B)) or reasoning-optimized models like DeepSeek-R1-Distill series. For Llama 3.3 (70B), CLONE consistently outperformed the best-performing baseline (i.e., 1-shot + CoT) across most metrics. Accuracy improved by 6.15%, while specificity increased by 10.71%. Precision and F1-score also saw notable gains, with precision rising by 6.55% and F1-score improving by 4.83%. These results demonstrate that the synthetic guideline enhances the diagnostic capabilities of larger LLMs by effectively guiding their clinical reasoning processes.

Similarly, for Phi 4 (14B), CLONE showed significant improvements compared to the highest-performing baseline (i.e., 0-shot + CoT). Accuracy increased by 6.16%, specificity by 14.29%, precision by 6.73%, and F1-score by 4.05%. These results further validate the efficacy of CLONE in enhancing diagnostic accuracy and interpretability, particularly for mid-sized LLMs.

In contrast, the results for Llama 3.1 (8B) were mixed. While specificity saw a substantial improvement of 42.86%, other metrics such as accuracy, sensitivity, and F1-score declined relative to the 1-shot + CoT baseline. Accuracy decreased by 12.31%, F1-score declined by 24.88%, and sensitivity dropped significantly by 54.06%. These results suggest that while the synthesized guideline can improve certain aspects of diagnostic reasoning, smaller models may lack the capacity to fully utilize the guideline, resulting in trade-offs in overall performance.

For the DeepSeek-R1-Distill models optimized for reasoning, the synthetic guideline did not consistently improve overall performance compared to the baselines. While certain metrics (e.g., specificity in the 14B model) showed modest gains under CLONE, improvements were not uniform across accuracy, sensitivity, precision, and F1-score. In most cases, accuracy even declined relative to the baseline settings. These mixed results suggest that the synthetic guideline does not necessarily offer additional benefits for all types of LLMs.

Overall, the findings indicate that CLONE enhances diagnostic accuracy and specificity for larger and mid-sized LLMs by effectively leveraging synthesized guidelines to improve clinical reasoning. However, its limited impact on smaller models and reasoning-optimized models suggests the need for further optimization to adapt the guidelines for less resource-intensive or specialized LLMs. Nevertheless, these results underscore the potential of LLM-driven synthetic guidelines to improve clinical decision-making in scenarios where larger models are feasible.

5.2 RQ2: How does an LLM-driven synthetic guideline impact on the quality of rationales?

Figure 3 illustrates that the CLONE framework outperformed the best-performing few-shot CoT baseline across all rationale quality criteria—*Consistency*, *Correctness*, *Specificity*, *Helpfulness*, and *Human-likeness*—for both Llama 3.3 (70B) and Phi 4 (14B). Llama 3.1 (8B) and DeepSeek-R1-Distill series were excluded from this evaluation due to its difficulty in generating accurate diagnoses.

For Llama 3.3, CLONE achieved winning rates of 83.1% in Consistency, 80.0% in Correctness, 86.2% in Specificity, 81.5% in Helpfulness, and 84.6% in Human-likeness, demonstrating its ability to generate rationales closely aligned with human reasoning. Similarly, Phi 4 achieved consistent improvements, with winning rates of 66.2%, 69.2%, 76.9%, 73.8%, and 73.8% across the same criteria. These results highlight the effectiveness of the synthesized guideline in producing rationales that are both reliable and interpretable, enhancing the clinical reasoning process. Overall, CLONE demonstrates its potential to address both accuracy and interpretability challenges in clinical applications.

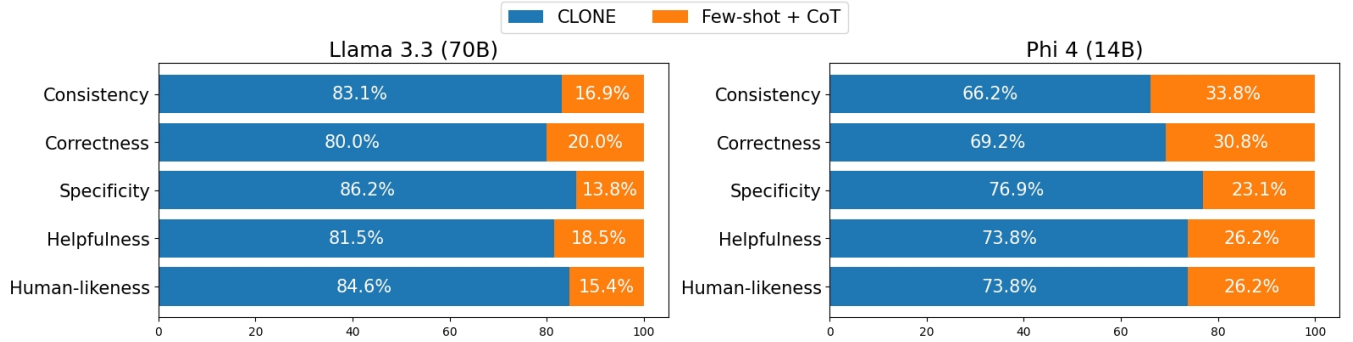
6 Discussion and Conclusion

In this study, we proposed and evaluated the CLONE framework, an LLM-driven synthetic guideline-based clinical reasoning approach designed to provide accurate diagnosis and human-readable rationales. Our findings demonstrate that the synthetic guideline significantly enhances both diagnostic accuracy and the quality of generated rationales. Specifically, the Llama 3.3 (70B) model achieved the highest performance, with an accuracy of 89.23%, surpassing the few-shot CoT baseline by 6.15%, emphasizing the effectiveness of the proposed framework. Moreover, the synthetic guideline contributed to the generation of rationales that were more consistent, correct, specific, helpful, and human-like compared to baseline approaches.

Regarding RQ1, the results indicate that the synthetic guideline improves diagnostic performance across most metrics, particularly in larger models. Notably, Llama 3.3 (70B) demonstrated notable improvements in accuracy, specificity, precision, and F1-score. Notably, specificity exhibited the most significant enhancement, with a 10.71% improvement over the baseline in Llama 3.3. This suggests that the synthetic guideline mitigates prediction biases observed in baseline approaches, which tended to overpredict MCI cases [19]. By providing structured and balanced guidance, the guideline reduced skewed predictions, resulting in more accurate and equitable classification for MCI diagnoses.

Table 1: Classification performance (%) on CLONE compared with baselines in MCI diagnosis across various LLMs. Bold represents the best and underlined represents the second-best performance for each model.

		Accuracy	Sensitivity	Specificity	Precision	F1-score
Llama 3.3 (70B)	0-shot + CoT	80.00	100.00	53.57	74.00	85.06
	1-shot + CoT	<u>83.08</u>	91.89	<u>71.43</u>	<u>80.95</u>	<u>86.08</u>
	2-shot + CoT	81.54	91.89	67.86	79.07	85.00
	CLONE	89.23	<u>94.59</u>	82.14	87.50	90.91
Phi 4 (14B)	0-shot + CoT	75.38	91.89	53.57	<u>72.34</u>	80.95
	1-shot + CoT	72.31	83.78	<u>57.14</u>	72.09	77.50
	2-shot + CoT	70.77	<u>86.49</u>	50.00	69.57	77.11
	CLONE	81.54	91.89	67.86	79.07	85.00
Llama 3.1 (8B)	0-shot + CoT	60.00	<u>89.19</u>	21.43	60.00	71.74
	1-shot + CoT	70.77	97.30	<u>35.71</u>	<u>66.67</u>	79.12
	2-shot + CoT	<u>63.08</u>	97.30	17.89	61.02	<u>75.00</u>
	CLONE	58.46	43.24	78.57	72.73	54.24
DeepSeek-R1-Distill (70B)	0-shot + CoT	<u>73.85</u>	100.00	19.05	72.13	83.81
	1-shot + CoT	75.38	<u>97.30</u>	46.43	<u>70.59</u>	<u>81.82</u>
	2-shot + CoT	69.23	94.59	<u>35.71</u>	66.04	77.78
	CLONE	60.00	100.00	7.14	58.73	74.00
DeepSeek-R1-Distill (32B)	0-shot + CoT	69.23	100.00	<u>4.76</u>	68.75	81.48
	1-shot + CoT	78.46	91.89	60.71	<u>75.56</u>	82.93
	2-shot + CoT	81.54	<u>97.30</u>	60.71	76.60	85.71
	CLONE	56.92	100.00	0.00	56.92	72.55
DeepSeek-R1-Distill (14B)	0-shot + CoT	80.00	90.91	57.14	81.63	86.02
	1-shot + CoT	<u>78.46</u>	<u>83.78</u>	<u>71.43</u>	<u>79.49</u>	<u>81.58</u>
	2-shot + CoT	72.31	81.08	60.71	73.17	76.92
	CLONE	73.85	72.97	75.00	79.41	76.06

**Figure 3: Winning rate of generated rationale quality between the proposed CLONE framework and the most accurate few-shot CoT baseline. The orange bars represent 1-shot CoT for Llama 3.3 (70B) and 0-shot CoT for Phi 4 (14B).**

For RQ2, the results reveal that the synthetic guideline-based approach enhances rationale quality by fostering step-by-step reasoning [5]. The CLONE framework leverages its synthesized unified guideline to ensure rationales are aligned with clinical reasoning standards by emulating human expert’s reasoning processes. This is particularly evident in addressing the limitations of baseline

methods, which often generate rationales lacking consistency or specificity. By offering a systematic and structured approach, the guideline enables LLMs to produce rationales that are both contextually appropriate and reflective of human-like reasoning patterns [31]. These strengths highlight the utility of the synthetic guideline in improving both diagnostic and the interpretability of

LLM-generated rationales, making them more reliable for clinical decision-making.

This study has several limitations that need to be addressed in future work. First, although the dataset *D* used for evaluation is clinically validated and based on real-world data, its relatively small size may limit the generalizability of the findings. Expanding the dataset in scale or modality could help capture a broader diversity of subject characteristics that are useful for identifying various signals of MCI. Furthermore, in real-world settings, it may be beneficial to build a pipeline that not only collects data but also continuously updates synthetic guidelines using the collected data to improve diagnostic performance. Second, this study evaluated only a limited range of open-source LLMs, which may not adequately represent the effectiveness of the proposed framework. Future work could validate the framework using more recent models which have powerful reasoning capability. Lastly, while the rubric-based evaluation method is quite robust, it relies on the subjective judgment of the LLM-as-Judge methodology [32], which may not fully capture nuanced human expectations in clinical reasoning. Incorporating additional validation layers, such as human expert evaluation, could enhance the reliability of the assessment and ensure closer alignment with expert judgment. Despite these limitations, this study makes significant contributions. The proposed CLONE framework demonstrates the potential of LLM-driven synthetic guidelines to enhance both diagnostic accuracy and rationale quality. This work represents a meaningful step toward developing interpretable and scalable LLM-driven tools for clinical applications. Furthermore, it highlights the broader applicability of synthetic guidelines in other domains requiring interpretable reasoning, paving the way for future advancements in leveraging LLMs for complex decision-making tasks.

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A Pseudo Code

Algorithm 1 CLONE

Input: Subset $D_{sub} = \{(s_j, r_j) | r_j = p(s_j)\}_{j=1}^n$, Neuropsychological test scores of target subject s^*

Output: Rationale R^* , Diagnosis D^*

Stage 1: Emulating Experts through Role-Playing

$\hat{A} \leftarrow []$

$A_e \leftarrow \{a_e^1, \dots, a_e^n\}$

▸ Initialize emulator agents

$Q \leftarrow \text{GetQuestionSet}()$

for $(S_j, R_j) \in D_{sub}$ **do in parallel do**

$a_j \leftarrow A_e[j].\text{Respond}(S_j, R_j, Q)$

▸ Collect response from emulator agents

$\hat{A}.\text{append}(a_j)$

end for

Stage 2: Synthesizing Step-by-Step Diagnostic Guidelines

$G_{draft} \leftarrow []$

for $a_j \in \hat{A}$ **do in parallel do**

$g_j \leftarrow A_g.\text{WriteDraft}(a_j)$

▸ Write draft guidelines

$G_{draft}.\text{append}(g_j)$

end for

$G \leftarrow A_g.\text{Consolidate}(G_{draft})$

▸ Write final unified guideline

Stage 3: Clinical Reasoning using the Guideline

$R^* \leftarrow A_p.\text{GenerateRationale}(s^*|G)$

▸ Perform clinical reasoning

$D^* \leftarrow A_p.\text{GenerateDiagnosis}(s^*, R^*|G)$

▸ Conduct diagnosis based on rationale

return R^*, D^*

B Example Prompts and Outputs

B.1 Stage 1: Emulating Experts through Role-Playing

Table 2: Example of prompts and emulator agent’s response in stage 1 of CLONE.

Role	Message
System	You are a professional neuropsychologist who has made the given diagnosis. Answer the user’s question.

Role	Message
User	### Subject's Neuropsychological Test Results #### Attention - Vigilance Test: Normal - ... #### Language - Comprehension: Normal - Repetition: 13 (%ile score: <5) - Short Form of the K-BNT (S-K-BNT): 10 (%ile score: 4.04, z-score: -1.75) #### Praxis - Ideomotor: 3 (%ile score: <5) #### Visuospatial Functions - Red Complex Figure Test (RCFT) - Copy Score: 36 (%ile score: 83.02, z-score: 0.96) - Copy Time (sec): 196 (%ile score: 33.83, z-score: -0.42) #### Memory - Seoul Verbal Learning Test - Elderly's Version (SVLT-E) - Immediate Recalls - Total Score: 23 (%ile score: 58.53, z-score: 0.22) - Delayed Recall: 7 (%ile score: 45.94, z-score: -0.10) - Recognition - ... #### Frontal / Executive Functions - ... - Korean Color Word Stroop Test 60 Seconds (K-CWST-60) - Word Reading (WR) - Correct: 112 (%ile score: 15≤) - Error: 0 (%ile score: 15≤) - ... #### Other Indexes - Geriatric Depression Scale (GDepS): 5 - Barthel-Activities of Daily Living (Barthel-ADL): 20 - Korean-Instrumental Activities of Daily Living (K-IADL): 0.00 ### Your Rationale and Diagnosis According to the neuropsychological assessment, the patient's general cognitive function was found to be good. The SNSB-C results suggested a decline in repetition ability and ideomotor apraxia. A decline in confrontational naming ability was indicated. The patient's visuospatial perception and analysis/construction abilities were at a very good level. Verbal memory was at an average level, and the patient's frontal lobe/executive functions were also considered to be maintained at a good level. The self-reported depression test did not suggest clinically significant levels of depression. Currently, the patient is reported to experience no difficulties in performing basic daily activities and complex daily functions involving the use of tools independently. The patient's current condition corresponds to non-amnestic Mild Cognitive Impairment (MCI). ### Question How did you interpret the test results? What were the decisive factors for the diagnosis?

Role	Message
Assistant (A_e)	To arrive at the diagnosis of non-amnesic Mild Cognitive Impairment (MCI), several key findings from the neuropsychological test results were considered. The decisive reasons for this diagnosis can be broken down into specific areas of cognitive function that were assessed: 1. **Language**: The patient showed a decline in repetition ability, as indicated by the low %ile score (<5) on the Repetition test and a significant impairment on the Short Form of the K-BNT (S-K-BNT) with a %ile score of 4.04. This suggests difficulties with language, specifically in aspects that do not primarily involve memory. 2. **Praxis**: The patient's performance on the Ideomotor test was also significantly impaired (<5 %ile), indicating ideomotor apraxia. This is a condition where an individual has difficulty in performing gestures or movements when asked, despite having the physical ability and desire to perform them. 3. **Visuospatial Functions**: In contrast to the impairments in language and praxis, the patient's visuospatial perception and analysis/construction abilities were found to be at a very good level, as evidenced by a high %ile score (83.02) on the Rey Complex Figure Test (RCFT): Copy Score. 4. **Memory**: The assessment of verbal memory using the Seoul Verbal Learning Test - Elderly's Version (SVLT-E) showed that the patient's immediate and delayed recall scores were at an average level, with %ile scores of 58.53 for total score and 45.94 for delayed recall. This suggests that while there might be some degree of memory impairment, it is not as pronounced as in amnesic MCI. 5. **Frontal/Executive Functions**: The patient's performance on tests assessing frontal lobe/executive functions was generally good. For example, the Korean Color Word Stroop Test 60 Seconds (K-CWST-60) and other executive function tests did not indicate significant impairments in this domain. 6. **Daily Functioning**: The patient reported no difficulties in performing basic daily activities and complex daily functions independently, as assessed by the Barthel Activities of Daily Living (Barthel-ADL) and Korean Instrumental Activities of Daily Living (K-IADL). 7. **Depression Screening**: The Geriatric Depression Scale (GDepS) and its short version (SGDepS) did not suggest clinically significant levels of depression, which is important to rule out as a potential cause of cognitive symptoms. Given these findings, the diagnosis of non-amnesic Mild Cognitive Impairment (MCI) is appropriate because it reflects the patient's specific pattern of cognitive impairments, primarily in language and praxis, without significant memory impairment or decline in daily functioning. This diagnosis suggests that while the patient exhibits some level of cognitive decline beyond what would be expected for their age and education, this decline does not meet the criteria for dementia, particularly given the preservation of independent daily functioning and the absence of significant amnesic symptoms (i.e., memory loss).

B.2 Stage 2: Synthesizing Step-by-step Diagnostic Guidelines

Table 3: Example of prompts and synthesized guideline in stage 2 of CLONE.

Role	Message
System	You are a professional neuropsychologist developing step-by-step guidelines on how to interpret SNSB-C test results to determine the subject's cognitive level.
User	Below is an explanation from a fellow neuropsychologist on why they interpreted the test results in that way and why they made such a diagnosis. Based on this, please create step-by-step guidelines for interpreting SNSB-C test results, focusing primarily on how to utilize objective statistical data. Do not use your own knowledge or experience; only use the information provided.
Assistant (A_g)	### Fellow Psychologist's Explanation: The answer set A from stage 1 is here <i>(...writing draft guidelines...)</i>
System	You are a professional neuropsychologist developing step-by-step guidelines on how to interpret SNSB-C test results to determine where the subject's cognitive level falls among the following options: - (A) Normal cognition or Subjective cognitive decline - (B) Mild cognitive impairment, Early stage of dementia, or Alzheimer's disease

Role	Message
User	Please consolidate the provided draft guidelines into a step-by-step interpretation of the SNSB-C test results. Do not use your own knowledge or experience, only the information provided. Organize the content in such a way that no examples from the draft guidelines are omitted. Include interpretations for both options (A) and (B) as mentioned above. Focus on using objective statistical data for interpretation. Specifically, the method for making the final diagnosis should be described in as much detail as possible, including many different cases. Now, please generate the step-by-step guidelines on how to interpret SNSB-C test results to determine the subject's cognitive level.
Assistant (A_g)	### Draft Guidelines: <i>The previously generated draft guidelines are here</i> ## Step-by-Step Guidelines for Interpreting SNSB-C Test Results ### Step 1: Evaluate General Cognitive Function - Review the total score to determine overall cognitive function. - Excellent performance (e.g., high %ile score and z-score) indicates strong general cognitive abilities, suggesting **Option A: Normal cognition or Subjective cognitive decline**. ### Step 2: Assess Attention and Concentration Abilities - Examine the Digit Span Test (DST) Forward and Backward scores, including %ile scores and z-scores. - Impaired performance (e.g., low %ile score ≤ 10 and z-score ≤ -1.0) may indicate difficulties with working memory and attention, suggesting **Option B: Mild cognitive impairment, Early stage of dementia, or Alzheimer's disease**. (...detailed guideline...) ### Step 10: Finalize Diagnosis - Based on the evaluation of all domains and consideration of mixed results, determine the final diagnosis: - **Option A: Normal cognition or Subjective cognitive decline**: If no significant cognitive impairment is observed, or if cognitive impairment is present but does not cause significant issues in daily life. - **Option B: Mild cognitive impairment, Early stage of dementia, or Alzheimer's disease**: If significant cognitive impairment is observed in one or more domains and causes significant issues in daily life.

B.3 Stage 3: Clinical Reasoning using the Guideline

Table 4: Example of prompts and generated clinical rationale with final diagnosis in stage 3 of CLONE.

Role	Message
System	You are a professional neuropsychologist tasked with deriving clinical rationale and utilizing it to diagnose the cognitive level of a subject.

Role	Message
User	### Task description 1. First, write the clinical rationale based on the given information below. 2. Next, use this rationale to make a final diagnosis. 3. Your response format should be as follows: ### Clinical Rationale [rationale] ### Diagnosis [(A)/(B)] <i>Examples for few-shot or the synthesized guideline are provided here</i> <i>The demographic information (e.g., age, gender, and years of education) and SNSB-C test scores of target subject are here</i> ### Question for diagnosis What is the diagnosis of this subject? - (A) Normal cognition or Subject cognitive decline. - (B) Mild cognitive impairment, Early stage of dementia, or Alzheimer's disease

Role	Message
Assistant (A_p)	### Clinical Rationale The subject, a 72-year-old female with 16 years of education, underwent a comprehensive neuropsychological assessment using the SNSB-C test battery. The evaluation aimed to assess various cognitive domains, including attention, language, visuospatial functions, memory, and frontal/executive functions. 1. **Attention**: The Digit Span Test (DST) Forward score was 5, with a %ile score of 13.57 and a z-score of -1.10, indicating some impairment in attention. However, the DST Backward score was 6, with a %ile score of 97.18 and a z-score of 1.91, suggesting relatively preserved working memory and attention abilities. 2. **Language**: The Short Form of the K-BNT (S-K-BNT) score was 11, with a %ile score of 17.94 and a z-score of -0.92, indicating some impairment in confrontation naming ability. The repetition test result showed a raw score of 13 but a %ile score < 5, suggesting difficulties with repetition. 3. **Visuospatial Functions**: The Rey Complex Figure Test (RCFT) Copy Score was 36, with a %ile score of 83.71 and a z-score of 0.98, indicating relatively preserved visuospatial perception and analysis/construction abilities. 4. **Memory**: The Seoul Verbal Learning Test-Elderly's Version (SVLT-E) Total Score was 22, with a %ile score of 58.07 and a z-score of 0.20. The Delayed Recall score was 5, with a %ile score of 22.43 and a z-score of -0.76, suggesting some impairment in memory. 5. **Frontal/Executive Functions**: The Digit Symbol Coding (DSC) test result showed a raw score of 87, with a %ile score of 99.63 and a z-score of 2.67, indicating relatively preserved psychomotor speed. The Controlled Oral Word Association Test (COWAT) animal category raw score was 17, with a %ile score of 53.58 and a z-score of 0.09, suggesting relatively preserved semantic generative naming ability. 6. **Other Domains**: The Geriatric Depression Scale (GDepS) score was 11, and the Short Version of the GDepS (SGDepS) score was 5, which does not indicate clinically significant levels of depression. The Barthel Activities of Daily Living (Barthel-ADL) score was 20, suggesting that the subject's ability to perform daily living activities independently is preserved. Considering these results, the subject exhibits some cognitive impairments in attention, language, and memory domains but also shows relatively preserved abilities in visuospatial functions, frontal/executive functions, and daily living activities. The presence of cognitive decline in one or more domains without significant impairment in daily life suggests a diagnosis of **Option A: Normal cognition or Subjective cognitive decline**. ### Diagnosis (A) Normal cognition or Subjective cognitive decline.

C Rubrics for LLM-as-Judge

Table 5: Rubrics to evaluate quality of generated rationales for clinical reasoning, adapted from [19].

Criteria	Description
Consistency	How much is a generated rationale not contradictory to the presented data and model prediction (or the ground-truth diagnosis)?
Correctness	How medically correct is the knowledge referred to in the rationale?
Specificity	How detailed and specific are the insights provided in the generated rationale?
Helpfulness	How much does a clinical rationale benefit the prediction towards the correct diagnosis?
Human-likeness	How well does a clinical rationale demonstrate the insight and understanding of the presented patient description or diagnosis in a way that matches the human behaviors?